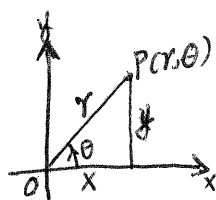
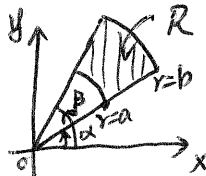


§15.4.



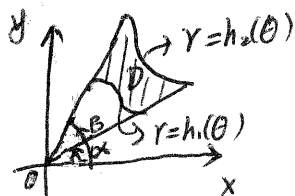
Polar coordinates (r, θ)
 $r^2 = x^2 + y^2, \tan \theta = \frac{y}{x}$
 $\Downarrow$
 $x = r \cos \theta, y = r \sin \theta$

Polar rectangle: $R = \{(r, \theta) \mid a \leq r \leq b, \alpha \leq \theta \leq \beta\}$.



$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) \underline{r dr d\theta}$$

General region: $D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$.



$$\iint_D f(x, y) dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) \underline{r dr d\theta}$$

The area of D is then
 $A(D) = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \underline{r dr d\theta} = \int_{\alpha}^{\beta} \frac{1}{2} (h_2^2(\theta) - h_1^2(\theta)) d\theta$

§15.6.

A rectangular box: $B = \{(x, y, z) \mid a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}$.

$$\iiint_B f(x, y, z) dV = \int_r^s \int_c^d \int_a^b f(x, y, z) dx dy dz$$

$$= \int_a^b \int_c^d \int_r^s f(x, y, z) dz dy dx$$

(change of integration order for x, y, z)

= ...

General bounded E in 3D space

TYPE 1: $E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$.

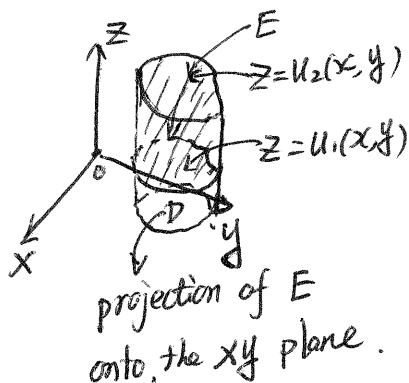
$$\text{then } \iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right] dA$$

$\hookrightarrow$ reduce to a double integral

Now consider D - plane region

I) $D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\} \Rightarrow$

$$\iiint_E f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dy dx$$



$$\text{II) } D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\} \Rightarrow$$

$$\iiint_E f(x, y, z) dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy.$$

TYPE 2: $E = \{(x, y, z) \mid (y, z) \in D, u_1(y, z) \leq x \leq u_2(y, z)\}$, then

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(y, z)}^{u_2(y, z)} f(x, y, z) dx \right] dA.$$

TYPE 3: $E = \{(x, y, z) \mid (x, z) \in D, u_1(x, z) \leq y \leq u_2(x, z)\}$, then

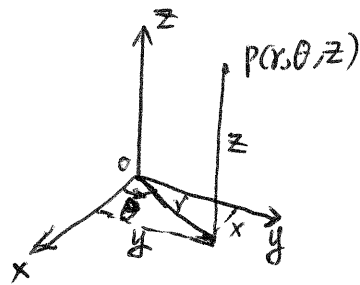
$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, z)}^{u_2(x, z)} f(x, y, z) dy \right] dA.$$

Volume of E: $V(E) = \iiint_E 1 dV = \iiint_E dV$

Let $\rho(x, y, z)$ be the density function of the object occupying E , then its mass is

$$m = \iiint_E \rho(x, y, z) dV$$

§15.7.



Cylindrical coordinates $(r, \theta, z) \Leftrightarrow \{\text{Polar coordinates}\} + \{z\}$

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}, \quad z = z$$

$\Leftrightarrow$

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Suppose E is a Type I region in the 3D space (§15.6)

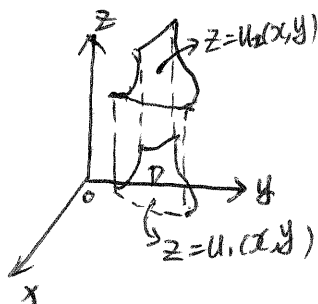
$$E = \{(x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$$

with

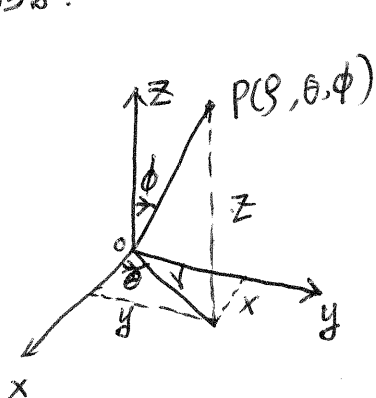
$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\} \rightarrow \text{polar coordinates}$$

then

$$\iiint_E f(x, y, z) dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) \underline{r dz dr d\theta}$$



8158.



spherical coordinate (ρ, θ, ϕ)

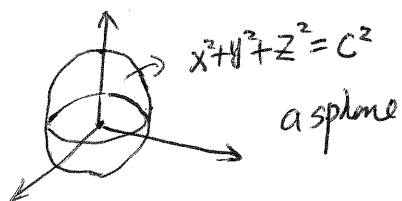
$$\rho \geq 0, 0 \leq \phi \leq \pi.$$

$$\rho = \sqrt{x^2 + y^2 + z^2}, \tan \theta = \frac{y}{x}, \cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

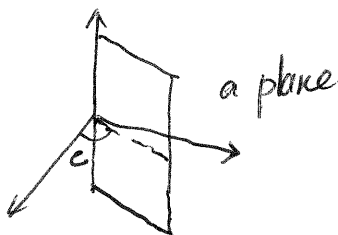


$$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$$

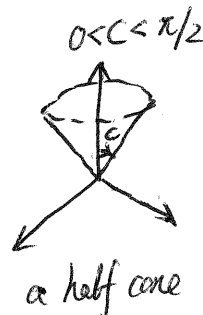
i) $\rho = C$ ($C > 0$)



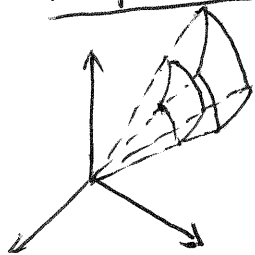
ii) $\theta = C$



iii) $\phi = C$



A spherical wedge $B = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b, \alpha \leq \theta \leq \beta, C \leq \phi \leq d\}$, then



$$\iiint_B f(x, y, z) dV = \int_C^d \int_\alpha^\beta \int_a^b f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi$$

General spherical region:

$$E = \{(\rho, \theta, \phi) \mid \alpha \leq \theta \leq \beta, C \leq \phi \leq d, g_1(\theta, \phi) \leq \rho \leq g_2(\theta, \phi)\}.$$

$$\Rightarrow \iiint_E f(x, y, z) dV = \int_C^d \int_\alpha^\beta \int_{g_1(\theta, \phi)}^{g_2(\theta, \phi)} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\theta d\phi$$

Chapter 16.

§16.1.

Vector fields

2D: $F(x, y) = \langle f_1(x, y), f_2(x, y) \rangle$ — two variables, output is a 2D vector

3D: $F(x, y, z) = \langle f_1(x, y, z), f_2(x, y, z), f_3(x, y, z) \rangle$
— three variables, output is a 3D vector.

Gravitational field:

$$F(\vec{x}) = \frac{mMG}{r^2} \left(-\frac{\vec{x}}{|\vec{x}|} \right) = -\frac{mMG}{|\vec{x}|^3} \vec{x} \quad \text{where } \vec{x} = (x, y, z).$$

Gradient field.

∇f , where f is a differentiable scalar function in $\mathbb{R}^d$.

$$d=2: \nabla f = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

$$d=3: \nabla f = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

Conservative Vector field

$$F = \nabla f$$

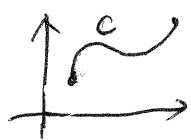
↳ potential function of the vector field F

↳ gradient field of f , is conservative ($\oint_C F \cdot d\mathbf{r} = 0$).

for example: The gravitational field is conservative since its potential function is $f(x, y, z) = \frac{mMG}{\sqrt{x^2 + y^2 + z^2}}$

§16.2.

Curve $C: x = x(t), y = y(t), a \leq t \leq b$



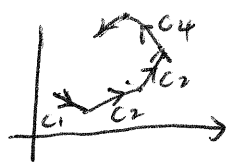
The line integral of the function f along C is

$$\int_C f(x, y) ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

If $f(x, y) = 1$, then it is the length of the curve

$$L = \int_C ds = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

Piece-wise smooth function $\{C_i\}_{i=1}^n = C$



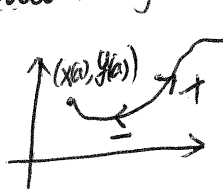
$$\int_C f(x,y) ds = \int_{C_1} f(x,y) ds + \dots + \int_{C_n} f(x,y) ds$$

The line integrals with respect to x and y .

$$\begin{cases} \int_C f(x,y) dx = \int_a^b f(x(t), y(t)) x'(t) dt \\ \int_C f(x,y) dy = \int_a^b f(x(t), y(t)) y'(t) dt \end{cases}$$

(Remark: they are dependant to the direction of C)

orientation of a curve: $x=x(t), y=y(t), a \leq t \leq b$



positive direction: increasing values of the parameter t .

$$\begin{cases} \int_{-C} f(x,y) dx = - \int_C f(x,y) dx \\ \int_{-C} f(x,y) dy = - \int_C f(x,y) dy \end{cases}$$

$$\text{but } \int_{-C} f(x,y) ds = \int_C f(x,y) ds$$

$\Delta x, \Delta y$ has signs

Δs is always positive.

Line integral in space

Curve $C: x=x(t), y=y(t), z=z(t), a \leq t \leq b$

$$\int_C f(x,y,z) ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b f(t) |r'(t)| dt$$

Line integrals of vector fields

vector field: $F(x,y,z)$

curve $C: r(t), a \leq t \leq b$

$$\boxed{\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt} = \int_C F \cdot \underbrace{T}_{\text{tangential vector } \frac{r'(t)}{|r'(t)|}} ds = \int_a^b F \cdot T |r'(t)| dt$$